MTH 9821 — American Options: Policy Evaluation and Fitted Q-Learning

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Equation numbers (1)–(13) refer to the assignment. Every number below is produced by `python src/run_all.py && python src/make_tables.py` (see README).

1. The value function

(1) Fixed-policy value. For a policy $a = (a_n)$ with $a_N \equiv 1$,

$$V_n^a(s,x) = \mathbb{E}\Big[\sum_{m=n}^{N} \beta^{m-n}\, x_m\, a_m(S_m)\, g(S_m) \;\Big|\; S_n = s,\ x_n = x\Big], \qquad x_{m+1} = (x_m - a_m(S_m))^+.$$

By (4), $x_m a_m(S_m) = 1$ only at the first exercise date $\geq n$ of an alive contract, so the sum is the single discounted payoff. Separating the $m = n$ term and conditioning on S_{n+1} (tower property; S is Markov and x_{n+1} is a known function of (s,x)):

$$V_n^a(s,x) = x\, a_n(s)\, g(s) + \beta\, \mathbb{E}\big[V_{n+1}^a\big(S_{n+1}, (x - a_n(s))^+\big) \,\big|\, S_n = s\big], \qquad V_N^a(s,x) = x\, g(s).$$

Exercised option. If $x = 0$ then $(0 - a_m)^+ = 0$, so $x_m = 0$ for all $m \geq n$, every term of the sum is zero and $V_n^a(s,0) = 0$: exercise pays once and ends the contract. *Function vs. path.* $V_n^a(s,x)$ is a deterministic number for each (s,x), a conditional mean. Along one simulated path, the realized discounted cash flow $\sum_m \beta^{m-n} x_m a_m g(S_m)$ is a random variable; its conditional mean is $V_n^a(s,x)$. Policy evaluation averages these realized cash flows.

(2) Bellman equation. With $x = 1$ the recursion gives two choices at date $n < N$. Exercise ($a_n = 1$) gives $g(s) + \beta\mathbb{E}[V_{n+1}(S_{n+1}, 0)] = g(s)$; continuing ($a_n = 0$) gives $\beta\mathbb{E}[V_{n+1}(S_{n+1}, 1) \mid S_n = s]$. Holding now and optimizing later defines

$$C_n^*(s) = \beta\, \mathbb{E}\big[V_{n+1}^*(S_{n+1}, 1) \,\big|\, S_n = s\big], \qquad V_n^*(s,1) = \max\{g(s),\, C_n^*(s)\} \quad (n < N),$$
$$V_N^*(s,1) = g(s), \qquad V_n^*(s,0) = 0.$$

Backward induction. Claim: for any admissible policy π, possibly using the whole observed history $\mathcal{F}_n$, the conditional expected remaining payoff of an alive contract is $\leq V_n^*(S_n, 1)$, with equality for $a_n^*(s) = \mathbf{1}\{g(s) \geq C_n^*(s)\}$. At $n = N$ settlement is forced, so equality holds. Assume the claim at $n+1$. If π exercises at n it receives $g(S_n) \leq V_n^*(S_n, 1)$. If it continues, its value is $\beta\mathbb{E}[\mathbb{E}[\text{payoff} \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] \leq \beta\mathbb{E}[V_{n+1}^*(S_{n+1}, 1) \mid \mathcal{F}_n] = C_n^*(S_n)$, where the last step uses the Markov property: given S_n, S_{n+1} is independent of the earlier history. So π gets at most $\max\{g, C_n^*\} = V_n^*$, and a^*, which picks the larger term at every date, attains it. Hence no policy, history-dependent or not, beats $V_0^*(S_0, 1)$.

(3) Two-period check. $q = (1.05 - 0.8)/(1.2 - 0.8) = 0.625$. Terminal payoffs: $S_{uu} = 144 \to 0$, $S_{ud} = S_{du} = 96 \to 4$, $S_{dd} = 64 \to 36$.

- Up ($S_1 = 120$): $C_1 = (0.625 \cdot 0 + 0.375 \cdot 4)/1.05 = 1.4286$, $g = 0 \Rightarrow$ continue, $V_1 = 1.4286$.
- Down ($S_1 = 80$): $C_1 = (0.625 \cdot 4 + 0.375 \cdot 36)/1.05 = 15.2381 < g = 20 \Rightarrow$ exercise, $V_1 = 20$.
- Inception: $C_0 = (0.625 \cdot 1.4286 + 0.375 \cdot 20)/1.05 = 7.9932 > g(100) = 0 \Rightarrow$ continue, $V_0 = 7.9932$.

Up–down and down–up both end at 96, but on down–up the contract was exercised at $S_1 = 80$, so $x_2 = 0$ and it receives nothing more; on up–down it is still alive ($x_2 = 1$) and pays 4 at t_2. The alive state depends on the path, not only on the current price.

2. Longstaff–Schwartz

(1)–(2) Implementation (`src/amop.py`, `train_lsm`). Planned index $\widehat{\tau} = N$ on every path. For $n = N-1, \ldots, 1$: form all responses (8), fit on in-the-money paths by `lstsq`, clip by (6), then set $\widehat{\tau} = n$ where (7) holds. At $n = 0$ the fitted value is $\max\{\overline{Y_0}, g(S_0)\}$ and the frozen decision uses the convention of (7). No fallback was triggered in any of the twelve fits.

(3) Questions. *Cash flow, not fitted value.* $Y_n^{(i)}$ is the payoff actually realized on path i by following the already fitted future policy. It is one noisy draw whose conditional mean given $S_n = s$ is the continuation value of that policy. Regression estimates this mean; no fitted value is fed back into the responses, so regression error affects later fits only through the decisions. *Future information.* Responses are training labels. The regression projects them onto functions of S_n only, so the fitted rule uses the current price. In evaluation the decision must be a stopping time; giving the policy future prices would be look-ahead (perfect foresight) and its payoff would not be attainable. *Discount to date n.* Rule (7) compares the response with $g(S_n)$, paid at t_n. Both must be in date-n money. Discounting to date 0 would shrink the continuation estimate by β^n and bias the rule toward exercise.

3. Fitted Q-learning

(1)–(2) Implementation (`train_fq`). $\widehat{V}_N = g$ on the training paths. For $n = N-1, \ldots, 1$: targets (11) from the frozen $\widehat{V}_{n+1}$, fit on all paths, clip by (6), then $\widehat{V}_n = \max\{g, \widehat{C}_n\}$ by (12). At $n = 0$: clipped average of (11) compared with $g(S_0)$. Actions then come from (7), and the frozen policy is evaluated on the fresh evaluation set (Part 4).

(3) Questions. *From (9) to (11).* $Q_n^*(s, \text{continue}) = \beta\mathbb{E}[V_{n+1}^*(S_{n+1}, 1) \mid S_n = s] = C_n^*(s)$. Replace the unknown V_{n+1}^* by the fitted $\widehat{V}_{n+1}$ and the conditional expectation by a least-squares regression on $\phi_d(S_n)$. Then $\beta\widehat{V}_{n+1}(S_{n+1}^{(i)})$ is a sample with conditional mean $\beta\mathbb{E}[\widehat{V}_{n+1}(S_{n+1}) \mid S_n = s]$, which is (11). *Contrast with (8).* In (11) only one transition S_{n+1} is realized and the rest is *estimated* ($\widehat{V}_{n+1}$). Its variance is low, but regression error and the upward bias of the max in (12) carry backward through all dates. In (8) the whole future cash flow is *realized* and only the decisions are estimated, so the variance is higher but no fitted value is propagated. *Out of the money.* With $s > K$ the stock can still fall below K before maturity, so $C_n^*(s) > 0$. In this recursion $\widehat{V}_n = \widehat{C}_n$ on such states, and those values become targets at date $n-1$ for every path that is out of the money at n. A fit on in-the-money states only would supply these values by extrapolating the polynomial outside its data (clipped to 0 or K, or badly wrong), and the error would enter all earlier targets. LSM does not have this problem: out of the money the decision is always "continue", and its responses are realized cash flows.

4. Comparison of frozen policies

Training seeds 17, 29, 43 (20,000 paths each, shared by both methods and degrees); one evaluation set, seed 20261008, 100,000 paths, used for every policy and never for tuning. Benchmark: finest tree value 6.07861 (change from previous refinement 0.00003).

(1) Checks and results. *Evaluator:* enumerating the four two-period paths with weights $q^2, q(1-q), (1-q)q, (1-q)^2$ and discount 1/1.05 gives 7.993197, equal to V_0 above (exercise dates: up–up and up–down at t_2, down–up and down–down at t_1). *European:* $\beta^N g(S_N)$ on the evaluation set gives 5.5876 (SE 0.0274) vs. Black–Scholes 5.5735; $z = 0.51$. *Reference:* an independent CRR tree (2500 steps, exercise every 50 steps) gives 6.0783717, equal to the supplied 2500-step value.

seed	method	d	fitted V_0	policy value	SE	95% CI	train (s)	eval (s)
17	LSM	2	5.9756	6.0589	0.0227	[6.0144, 6.1033]	0.18	0.38
17	FQ	2	7.8539	5.8516	0.0264	[5.7998, 5.9035]	0.15	0.38
17	LSM	4	5.9903	6.0780	0.0226	[6.0337, 6.1223]	0.22	0.34
17	FQ	4	6.9007	6.0191	0.0243	[5.9716, 6.0667]	0.15	0.38
29	LSM	2	6.0806	6.0518	0.0224	[6.0079, 6.0958]	0.19	0.48
29	FQ	2	7.9505	5.8505	0.0265	[5.7985, 5.9025]	0.18	0.43
29	LSM	4	6.1011	6.0699	0.0225	[6.0259, 6.1139]	0.22	0.44
29	FQ	4	6.9897	6.0050	0.0246	[5.9568, 6.0532]	0.23	0.48
43	LSM	2	6.0950	6.0560	0.0229	[6.0111, 6.1010]	0.22	0.39
43	FQ	2	7.9321	5.8494	0.0265	[5.7974, 5.9014]	0.21	0.49
43	LSM	4	6.1032	6.0774	0.0227	[6.0329, 6.1219]	0.31	0.55
43	FQ	4	7.0090	5.9778	0.0248	[5.9291, 6.0265]	0.23	0.41

Table 1: Twelve frozen policies. Policy value, SE and CI from (13) on the evaluation set. Times exclude path simulation (≈0.04 s per training set, 0.21 s for the evaluation set; see `results.csv`).

Discrepancies. (i) FQ fitted values (6.90–7.95) exceed their policy values by 0.9–2.1, roughly 35–80 SE: this is not Monte Carlo error. Each date's fitted value inherits the regression error of all later dates, and the max in (12) turns that error into an upward bias. A degree-2 polynomial fitted on all paths also places $\widehat{C}_n$ above g in the region where exercise is optimal, so the policy exercises too late. Raising the degree to 4 lowers the fitted value by about 0.95 and raises the policy value by 0.13–0.17, consistent with approximation error. (ii) LSM fitted values differ from policy values by -0.08 to $+0.04$. They are averages over only 20,000 training paths (SE ≈ 0.05), so these gaps are within about 1.6 training SE. (iii) All twelve policy values lie below the benchmark; LSM $d = 4$ is within 0.009 of it for every seed.

(2) Paired comparisons (FQ − LSM on common evaluation paths, (13) on the differences).

seed	d	FQ − LSM	SE	95% CI
17	2	-0.2072	0.0149	[-0.2365, -0.1779]
17	4	-0.0589	0.0094	[-0.0773, -0.0404]
29	2	-0.2013	0.0154	[-0.2315, -0.1711]
29	4	-0.0649	0.0106	[-0.0856, -0.0442]
43	2	-0.2066	0.0147	[-0.2355, -0.1777]
43	4	-0.0996	0.0109	[-0.1210, -0.0782]

Table 2: FQ − LSM by seed and degree (`paired.csv`).

All six FQ − LSM intervals lie below zero: LSM gives the better policy for every seed and degree. The gap is about -0.20 at $d = 2$ and -0.06 to -0.10 at $d = 4$. Pairing removes the common path noise: the paired SE (0.009–0.015) is well below the ≈ 0.034 an unpaired difference would have. Degree 4 narrows the FQ − LSM gap for every seed. From Table 1, degree 4 raises the policy value for both methods: LSM from 6.052–6.059 to 6.070–6.078, FQ from 5.849–5.852 to 5.978–6.019. Across seeds, the LSM $d = 4$ policy values range over 6.070–6.078 and FQ $d = 4$ over 5.978–6.019, so FQ is also more sensitive to the training sample. Three seeds are too few to estimate this spread precisely, and the twelve prices are not pooled.

(3) Decisions at $n = 25$, seed 17.

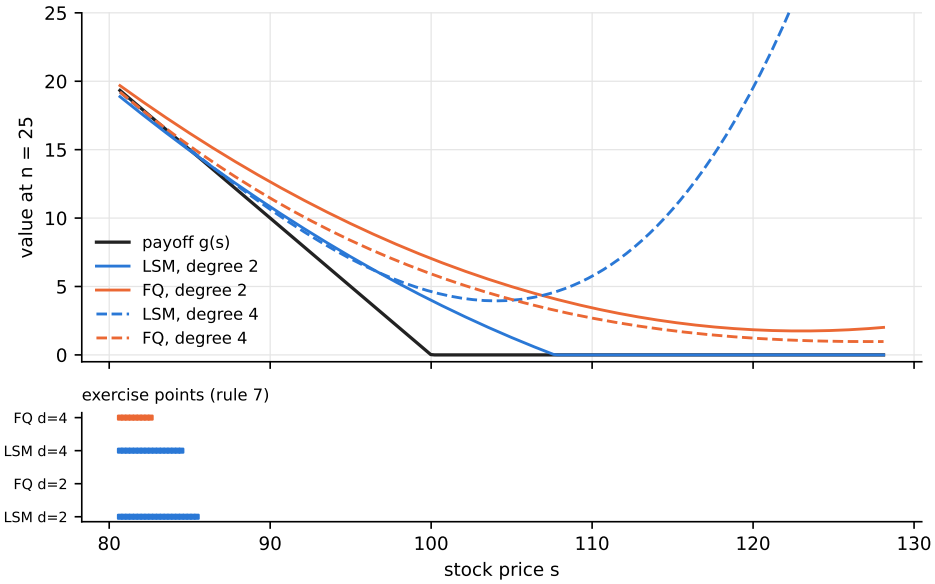


Figure 1: Clipped continuation $\widehat{C}_{25}$ and g on 200 spots between the 5th and 95th percentiles of the date-25 training spots (80.7–128.1). Bottom: grid points where (7) exercises.

The exercise regions were computed point by point. On this grid each is a single interval: LSM $d = 2$ exercises for $s \leq 85.4$, LSM $d = 4$ for $s \leq 84.5$, FQ $d = 4$ for $s \leq 82.6$, and FQ $d = 2$ never exercises ($\widehat{C}_{25} > g$ everywhere). The LSM $d = 4$ curve rises sharply out of the money because it is fitted on in-the-money paths only; there $g = 0$, so the policy continues anyway. *More flexible fit:* yes, the independently evaluated price improves with degree 4 for both methods (Table 1), a lot for FQ and slightly for LSM. *Small price impact near a crossing:* where $g(s) \approx C_n^*(s)$, choosing the wrong action loses only $|g(s) - C_n^*(s)| \approx 0$. A small shift of the boundary therefore costs little: the LSM $d = 2$ and $d = 4$ boundaries differ by about 1 in s, yet their prices differ by only about 0.02.

(4) Interpretation. Conditional on the training data D, a frozen policy is a fixed rule using only (n, S_n), and the evaluation paths are independent of D. It is therefore admissible, and by Part 1(2) $\mathbb{E}[H \mid D] = V_0^{\hat{a}}(S_0, 1) \leq V_0^*(S_0, 1) \approx 6.0786$. *Test means* estimate $\mathbb{E}[H \mid D]$ with error of order one SE (≈ 0.023), so a policy close to optimal can produce a sample mean above V_0^* by chance. *Fitted values* are not payoffs of an admissible policy. LSM's is an in-sample average on the same paths used to choose the decisions (optimistic, plus training noise). FQ's is built from fitted values with an upward max bias and basis error. Neither has to lie below V_0^*, and FQ's do not. *What the SEs measure:* only evaluation Monte Carlo noise for a fixed policy. They omit: training-sample variation (a different seed gives a different policy; compare the seed spread), approximation error from the polynomial basis and degree, and reference error (finite tree; the last refinement change 0.00003 is an empirical check, not a bound).

Use of AI assistance

We used Claude (Anthropic) as our AI assistant for derivation checks, code drafting and debugging, and report and slide drafting. We checked the work independently of it: the two-period model by hand and by path enumeration through the shared evaluator; the European price against Black–Scholes; the reference with our own CRR tree; the vectorized evaluator against a path-by-path loop version (maximum difference 4×10^{-15}); and the effect of the degree on both fitted and policy values. Both partners reviewed the derivations and the code and can explain them.